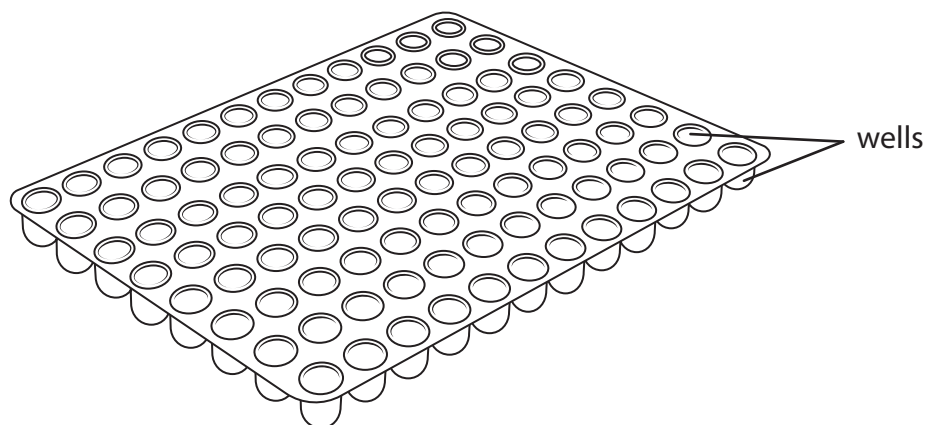


5 Antimicrobial substances can be tested using a Minimum Inhibitory Concentration (MIC) assay.

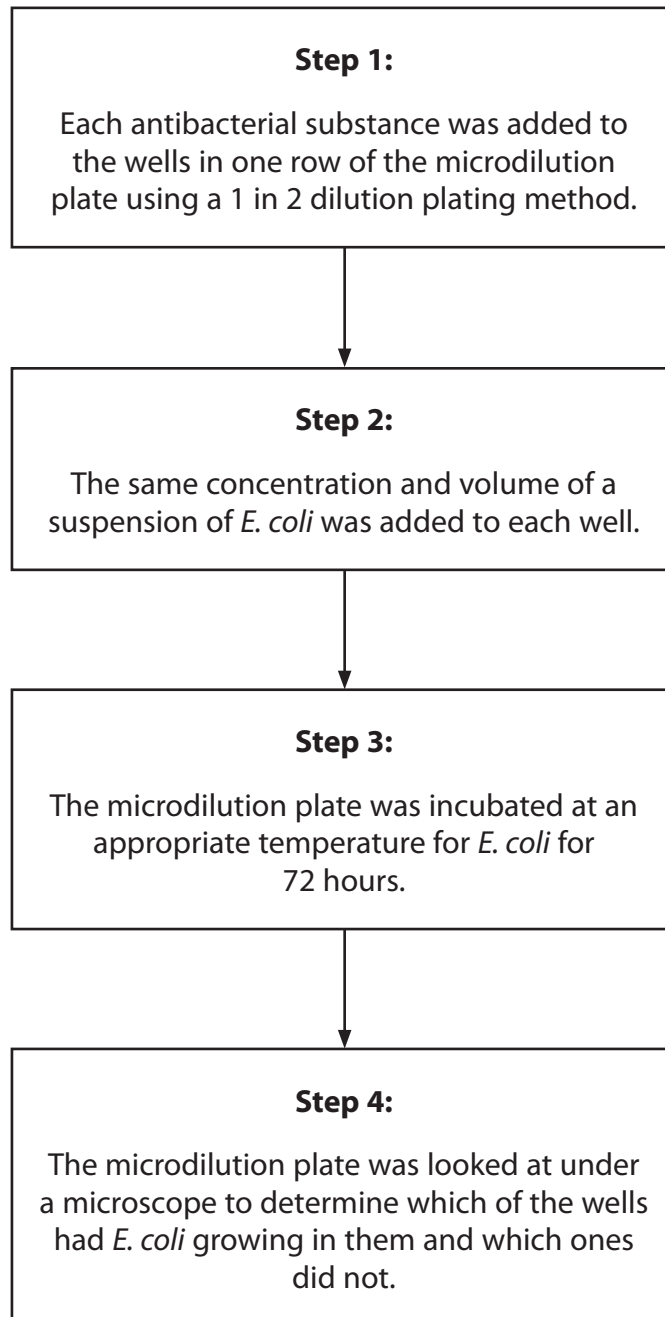
This assay determines the lowest concentration of an antimicrobial substance that prevents visible growth of bacteria.

A microdilution plate is used in these assays. It is made of plastic and contains small wells that the antimicrobial substance and the bacteria can be added to.

The diagram shows a microdilution plate.



An investigation tested eight antimicrobial substances on one type of bacteria, *E. coli*.
Appropriate controls were included in this investigation.
The diagram shows the steps involved.



(a) All steps in this investigation had to be carried out using aseptic technique.

(i) State the meaning of the term **aseptic technique**.

(1)

(ii) Describe two aseptic techniques that could be used in this investigation.

(2)

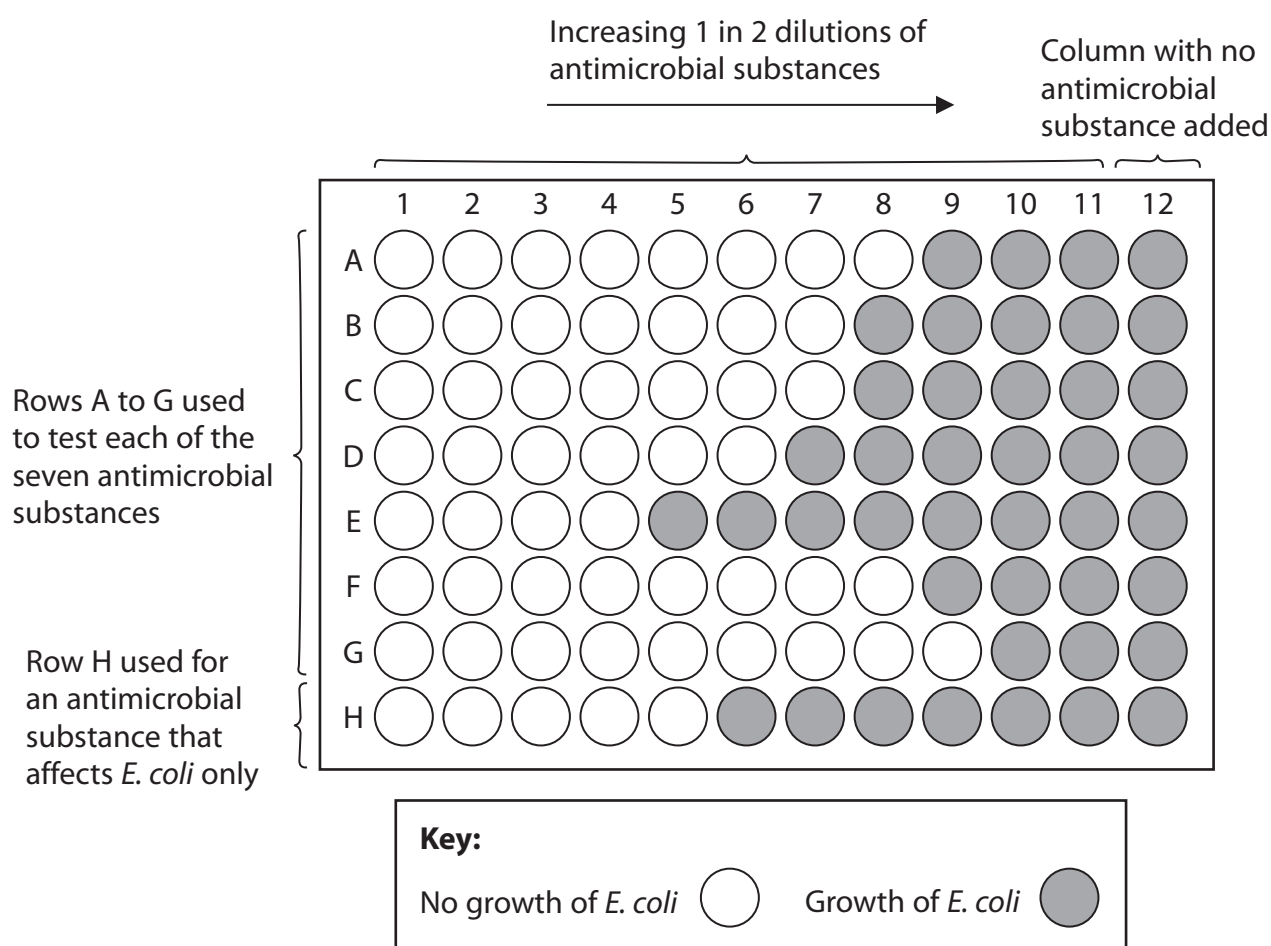
(iii) Explain why using aseptic technique in this investigation is important.

(2)

(b) Explain why the microdilution plate had to be incubated at an appropriate temperature for *E. coli* for 72 hours.

(2)

(c) The diagram shows the results of the MIC assay from this investigation.



- (i) Explain why an antimicrobial substance that affects *E. coli* only was included in this assay (row **H**).

(2)

- (ii) Explain why there was one column that had no antimicrobial substance added to it (column 12).

(2)

- (iii) The MIC for the antimicrobial substance used in row **E** was in column 4.

Describe how a 1 in 2 dilution plating method would have been carried out to achieve the dilution in this well.

(2)

(iv) Calculate how many times more effective the antimicrobial substance used in row **G** is than the antimicrobial substance used in row **E**.

(2)

Answer

(Total for Question 5 = 15 marks)